The Æther-flow Interpretation of Relativity

Substrate-Response / Self-Response Charge-Polarization Candidate
Exact-Preservation Derivation-Algebra / Balance-Law / Equilibrium-Manifold
Origin Note

Abstract

The exact-preservation charge-generator / continuity-Hessian deeper-origin note already shows that the previously recorded charge-generator / continuity-Hessian layer is not primitive at present scope. The recorded charge Lie algebras arise as faithful visible quotients of exact-preserving derivation algebras, the recorded density readouts and continuity operators arise as linearizations of primitive nonlinear balance laws, and the recorded ambient equilibrium Hessians arise as second variations of primitive equilibrium energies with tangent-kernel and positive-gap structure forced by equilibrium-manifold geometry. What remained open is one layer deeper again: why those exact-preserving derivation algebras, primitive nonlinear density/balance laws, and primitive equilibrium energies/manifolds themselves should exist at all rather than being the present deepest disciplined postulate.

The present note addresses exactly that narrower burden. It keeps fixed the same exact-preservation normal form, the same carrier-origin functional, the same primitive reservoir, the same microscopic-equilibrium completion reservoir, the same Σ^{cmp} solve, the same $\Pi_{\mathrm{free}} H = 0$ outcome, the same one operative metric, and the same recorded Phase D / E and Phase F gains. Its positive move is to place the recorded derivation-law / balance-law / equilibrium-manifold layer under still more primitive substrate structure:

1. finite-dimensional exact-preserving reconfiguration Lie groups with bi-invariant metrics and exact-preserving unitary actions on the sharper sectors, whose visible tangent actions are exactly the recorded exact-preserving derivation algebras;
2. raw substrate transport manifolds with density observables, stationary transport fields, and flux operators, whose pullbacks along exact-preserving visible embeddings are exactly the recorded primitive nonlinear density and balance laws;
3. raw equilibrium-reduction manifolds with exact-preserving reduction maps, deeper equilibrium energies, and equilibrium submanifolds, whose pullbacks are exactly the recorded primitive ambient equilibrium energies and manifolds.

At current exact-preservation scope, the derivation layer is therefore no longer primitive. The recorded derivation algebras

$$\mathfrak{d}_{\mathrm{car}}, \qquad \mathfrak{d}_{\mathrm{cur}}, \qquad \mathfrak{d}_{\mathrm{hom}}$$

are the visible tangent quotients of deeper exact-preserving reconfiguration groups. The primitive density and balance laws are no longer primitive either. The recorded nonlinear density maps

$$\widehat{\mathcal{Q}}_{\mathrm{car}}, \qquad \widehat{\mathcal{Q}}_{\mathrm{cur}}, \qquad \widehat{\mathcal{Q}}_{\mathrm{hom}}$$

and balance defects

$$\widehat{\mathfrak{c}}_{\mathrm{car}}, \qquad \widehat{\mathfrak{c}}_{\mathrm{cur}}, \qquad \widehat{\mathfrak{c}}_{\mathrm{hom}}$$

are the pullbacks of raw substrate transport laws, and their linearizations again force the recorded density readouts

$$\mathcal{Q}_{\mathrm{car}}, \qquad \mathcal{Q}_{\mathrm{cur}}, \qquad \mathcal{Q}_{\mathrm{hom}}$$

and continuity operators

$$\partial_{\mathrm{car}}, \qquad \partial_{\mathrm{cur}}, \qquad \partial_{\mathrm{hom}}.$$

Likewise the primitive ambient equilibrium energies

$$\mathcal{U}_{\mathrm{car}}, \qquad \mathcal{U}_{\mathrm{cur}}, \qquad \mathcal{U}_{\mathrm{hom}}$$

and equilibrium manifolds

$$\mathcal{N}_{\mathrm{car}}, \qquad \mathcal{N}_{\mathrm{cur}}, \qquad \mathcal{N}_{\mathrm{hom}}$$

are recovered as reduced pullbacks of deeper equilibrium geometry. Their Hessians are again the recorded ambient equilibrium Hessians, their kernel projectors are again the tangent projectors onto the equilibrium manifolds, and their positive gaps are again forced by strict transverse coercivity.

Because the present note recovers exactly the same derivation-law / balance-law / equilibrium-manifold package, it recovers exactly the same charge-generator / continuity-Hessian layer and hence exactly the same symmetry-charge / compatibility-response functional

$$\mathscr{F}_{\mathrm{sel}}.$$

Therefore the same Hessian-layer functional, the same microphysical exact-preservation functional, the same exact-preservation normal form, the same carrier-origin functional, the same primitive reservoir, the same microscopic-equilibrium reservoir, the same $\Pi_{\mathrm{free}} H = 0$ outcome, the same Σ^{cmp} solve, the same one operative metric, and the same recorded Phase D / E and Phase F gains remain fixed.

The scope remains disciplined. This note does not claim a final microscopic completion, a unique ultimate substrate theory, or a completed first-principles GR derivation. Its exact positive claim is narrower: the current derivation-law / balance-law / equilibrium-manifold layer is itself no longer primitive at present scope, but is forced by a still more primitive reconfiguration-group / transport-law / equilibrium-reduction substrate layer while preserving the same downstream package.

1. Introduction

The active one-metric charge-polarization line has now reached the point at which the charge-generator / continuity-Hessian layer is already recorded as the visible image of a deeper derivation-law / balance-law / equilibrium-manifold construction. The next burden therefore narrows again. It is not another packaging pass. It is not stronger promotion language. It is not, by default, another move onto the anchored positive-pair side line. It is to go one layer below the newly recorded derivation / balance / equilibrium construction itself and explain why *that* layer is forced.

That burden has four parts. First, one must explain why the exact-preserving derivation algebras

$$\mathfrak{d}_{\mathrm{car}}, \qquad \mathfrak{d}_{\mathrm{cur}}, \qquad \mathfrak{d}_{\mathrm{hom}}$$

should exist rather than being inserted by hand as the present deepest disciplined generators.

Second, one must explain why the primitive nonlinear density and balance laws

$$\widehat{\mathcal{Q}}_{\mathrm{car}}, \qquad \widehat{\mathcal{Q}}_{\mathrm{cur}}, \qquad \widehat{\mathcal{Q}}_{\mathrm{hom}}, \qquad \widehat{\mathfrak{c}}_{\mathrm{car}}, \qquad \widehat{\mathfrak{c}}_{\mathrm{cur}}, \qquad \widehat{\mathfrak{c}}_{\mathrm{hom}}$$

should arise from yet more primitive substrate transport structure, and why their linearizations should again yield the recorded density readouts, continuity operators, and compatibility-current maps. Third, one must explain why the primitive ambient equilibrium energies

$$\mathcal{U}_{\mathrm{car}}, \qquad \mathcal{U}_{\mathrm{cur}}, \qquad \mathcal{U}_{\mathrm{hom}}$$

and equilibrium manifolds

$$\mathcal{N}_{\mathrm{car}}, \qquad \mathcal{N}_{\mathrm{cur}}, \qquad \mathcal{N}_{\mathrm{hom}}$$

should arise from yet more primitive equilibrium geometry, with tangent-equilibrium kernel projectors and strict transverse coercivity again forcing the ambient Hessians and their positive gaps. Fourth, one must re-show that solving those deeper burdens changes none of the already recorded gains: the same $\mathscr{F}_{\mathrm{sel}}$, the same exact-preservation normal form, the same carrier-origin functional, the same primitive and microscopic-equilibrium reservoirs, the same Σ^{cmp} solve, the same $\Pi_{\mathrm{free}} H = 0$ outcome, the same one operative metric, and the same recorded Phase D / E and Phase F gains must remain intact.

The key move of the present note is to replace the primitive postulation of the current derivation-law / balance-law / equilibrium-manifold layer by three still more primitive substrate ingredients on the same sharper sectors. First come exact-preserving reconfiguration groups acting unitarily on those sectors. Their visible tangent quotients are exactly the recorded derivation algebras. Second come raw substrate transport manifolds with density observables, transport fields, and flux operators. Pulling those laws back along exact-preserving visible embeddings gives exactly the recorded primitive density and balance layer. Third come raw equilibrium-reduction manifolds carrying deeper equilibrium energies and equilibrium submanifolds. Pulling those energies back along exact-preserving reduction maps gives exactly the recorded primitive ambient energies and equilibrium manifolds, so the same Hessians, tangent kernel projectors, and positive gaps follow again. Because the previous note already proves that every later stage depends only on the induced derivation / balance / equilibrium package, preserving that package preserves the full downstream chain.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact relativistic theory statement in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active manuscript sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim a final microscopic

completion, a uniqueness theorem for every conceivable deeper substrate realization, or a completed GR derivation from first principles. Its narrower exact claim is only that the current derivation-law / balance-law / equilibrium-manifold layer itself arises from a still more primitive reconfiguration-group / transport-law / equilibrium-reduction substrate layer, with no change to the already recorded downstream package.

2. Fixed Inputs from the Recorded Completion Line

Definition 1 (Fixed branch and downstream completion target)**.** *The present note keeps fixed the same charge-polarization branch*

$$g^{\mathrm{cand}} = g^{\mathrm{br}} + \Sigma^{\mathrm{IER}} + \Sigma^{\mathrm{resp}}, \qquad g^{\sharp} = g^{\mathrm{br}} + \Sigma^{\mathrm{IER}} + \Sigma^{\mathrm{resp}} + \Sigma^{\mathrm{cmp}}, \tag{1}$$

with lifted variables satisfying

$$\Xi^{*}\widehat{\Sigma}^{\mathrm{IER}} = \Sigma^{\mathrm{IER}}, \qquad \Xi^{*}\widehat{\Sigma}^{\mathrm{resp}} = \Sigma^{\mathrm{resp}}, \qquad \Xi^{*}\widehat{\Sigma}^{\mathrm{cmp}} = \Sigma^{\mathrm{cmp}}. \tag{2}$$

The same completion class and free-sector verdict remain fixed:

$$\widehat{\Sigma}^{\mathrm{cmp}} \in \mathscr{C}_{\mathrm{cmp}}, \qquad \Pi_{\mathrm{free}}\widehat{\Sigma}^{\mathrm{cmp}} = 0. \tag{3}$$

The same substrate and observer completion solves remain fixed as well:

$$\widehat{\mathcal{L}}(\widehat{\Sigma}^{\mathrm{cmp}}) = -\widehat{\mathcal{V}}^{\mathrm{res}}, \qquad \mathcal{L}_{g^{\mathrm{br}}}(\Sigma^{\mathrm{cmp}}) = -\mathcal{V}^{\mathrm{res}}_{\mu\nu}. \tag{4}$$

Any deeper note is admissible here only if it preserves (1)–(4) *exactly.*

Definition 2 (Recorded derivation-law / balance-law / equilibrium-manifold layer)**.** *The current charge-generator / continuity-Hessian deeper-origin note fixes sharper Hilbert sectors*

$$\mathscr{U}_{\mathrm{car}}, \qquad \mathscr{U}_{\mathrm{cur}}, \qquad \mathscr{U}_{\mathrm{hom}}$$

equipped with finite-dimensional exact-preserving derivation algebras

$$\mathfrak{d}_{\mathrm{car}} \subset \mathfrak{u}(\mathscr{U}_{\mathrm{car}}), \qquad \mathfrak{d}_{\mathrm{cur}} \subset \mathfrak{u}(\mathscr{U}_{\mathrm{cur}}), \qquad \mathfrak{d}_{\mathrm{hom}} \subset \mathfrak{u}(\mathscr{U}_{\mathrm{hom}}),$$

closed under commutators and carrying positive-definite Ad*-invariant inner products. Their action-null ideals are*

$$\mathfrak{n}_{\mathrm{car}} \equiv \{X \in \mathfrak{d}_{\mathrm{car}} : X\mathfrak{U} = 0 \textit{ for all } \mathfrak{U} \in \mathscr{U}_{\mathrm{car}}\}, \tag{5}$$

$$\mathfrak{n}_{\mathrm{cur}} \equiv \{Y \in \mathfrak{d}_{\mathrm{cur}} : Y\mathfrak{V} = 0 \textit{ for all } \mathfrak{V} \in \mathscr{U}_{\mathrm{cur}}\}, \tag{6}$$

$$\mathfrak{n}_{\mathrm{hom}} \equiv \{Z \in \mathfrak{d}_{\mathrm{hom}} : Z\mathfrak{Q} = 0 \textit{ for all } \mathfrak{Q} \in \mathscr{U}_{\mathrm{hom}}\}, \tag{7}$$

and the recorded charge Lie algebras are the visible quotients

$$\mathfrak{g}_{\mathrm{car}} \equiv \mathfrak{d}_{\mathrm{car}}/\mathfrak{n}_{\mathrm{car}}, \qquad \mathfrak{g}_{\mathrm{cur}} \equiv \mathfrak{d}_{\mathrm{cur}}/\mathfrak{n}_{\mathrm{cur}}, \qquad \mathfrak{g}_{\mathrm{hom}} \equiv \mathfrak{d}_{\mathrm{hom}}/\mathfrak{n}_{\mathrm{hom}}. \tag{8}$$

Their induced faithful skew-adjoint generator maps are

$$\Gamma_{\mathrm{car}}([X])\mathfrak{U} \equiv X\mathfrak{U}, \qquad \Gamma_{\mathrm{cur}}([Y])\mathfrak{V} \equiv Y\mathfrak{V}, \qquad \Gamma_{\mathrm{hom}}([Z])\mathfrak{Q} \equiv Z\mathfrak{Q}. \tag{9}$$

Compact Lie integration yields the recorded symmetry actions

$$\mathscr{U}_{\mathrm{car}} : G_{\mathrm{car}} \to \mathcal{U}(\mathscr{U}_{\mathrm{car}}), \qquad \mathscr{U}_{\mathrm{cur}} : G_{\mathrm{cur}} \to \mathcal{U}(\mathscr{U}_{\mathrm{cur}}), \qquad \mathscr{U}_{\mathrm{hom}} : G_{\mathrm{hom}} \to \mathcal{U}(\mathscr{U}_{\mathrm{hom}}),$$

their Haar-average orthogonal projectors

$$\Pi_{\mathrm{sym,car}}, \qquad \Pi_{\mathrm{sym,cur}}, \qquad \Pi_{\mathrm{sym,hom}},$$

and the same Casimir operators

$$\mathbb{C}_{\mathrm{car}}, \qquad \mathbb{C}_{\mathrm{cur}}, \qquad \mathbb{C}_{\mathrm{hom}}.$$

On the symmetry-fixed sectors

$$\operatorname{ran}\Pi_{\mathrm{sym,car}}, \qquad \operatorname{ran}\Pi_{\mathrm{sym,cur}}, \qquad \operatorname{ran}\Pi_{\mathrm{sym,hom}},$$

the same note fixes Fréchet-smooth exact-preserving nonlinear density maps

$$\widehat{\mathcal{Q}}_{\mathrm{car}} : \operatorname{ran}\Pi_{\mathrm{sym,car}} \to \mathscr{N}_{\mathrm{car}}, \qquad \widehat{\mathcal{Q}}_{\mathrm{cur}} : \operatorname{ran}\Pi_{\mathrm{sym,cur}} \to \mathscr{N}_{\mathrm{cur}}, \qquad \widehat{\mathcal{Q}}_{\mathrm{hom}} : \operatorname{ran}\Pi_{\mathrm{sym,hom}} \to \mathscr{N}_{\mathrm{hom}},$$

and Fréchet-smooth exact-preserving balance defects

$$\widehat{\mathfrak{c}}_{\mathrm{car}} : \operatorname{ran}\Pi_{\mathrm{sym,car}} \to \mathscr{M}_{\mathrm{car}}, \qquad \widehat{\mathfrak{c}}_{\mathrm{cur}} : \operatorname{ran}\Pi_{\mathrm{sym,cur}} \to \mathscr{M}_{\mathrm{cur}}, \qquad \widehat{\mathfrak{c}}_{\mathrm{hom}} : \operatorname{ran}\Pi_{\mathrm{sym,hom}} \to \mathscr{M}_{\mathrm{hom}}.$$

Their linear density readouts are

$$\mathcal{Q}_{\mathrm{car}} \equiv D\widehat{\mathcal{Q}}_{\mathrm{car}}(0), \qquad \mathcal{Q}_{\mathrm{cur}} \equiv D\widehat{\mathcal{Q}}_{\mathrm{cur}}(0), \qquad \mathcal{Q}_{\mathrm{hom}} \equiv D\widehat{\mathcal{Q}}_{\mathrm{hom}}(0), \tag{10}$$

and the recorded continuity operators are the unique bounded maps on the closed visible density ranges satisfying

$$D\widehat{\mathfrak{c}}_{\mathrm{car}}(0) = \partial_{\mathrm{car}}\mathcal{Q}_{\mathrm{car}}, \qquad D\widehat{\mathfrak{c}}_{\mathrm{cur}}(0) = \partial_{\mathrm{cur}}\mathcal{Q}_{\mathrm{cur}}, \qquad D\widehat{\mathfrak{c}}_{\mathrm{hom}}(0) = \partial_{\mathrm{hom}}\mathcal{Q}_{\mathrm{hom}}. \tag{11}$$

The recorded compatibility-current maps are identified with the balance defects:

$$\mathfrak{c}_{\mathrm{car}} \equiv \widehat{\mathfrak{c}}_{\mathrm{car}}, \qquad \mathfrak{c}_{\mathrm{cur}} \equiv \widehat{\mathfrak{c}}_{\mathrm{cur}}, \qquad \mathfrak{c}_{\mathrm{hom}} \equiv \widehat{\mathfrak{c}}_{\mathrm{hom}}. \tag{12}$$

Hence the compatibility operators are again

$$\mathcal{C}_{\mathrm{car}} = \partial_{\mathrm{car}}\mathcal{Q}_{\mathrm{car}}, \qquad \mathcal{C}_{\mathrm{cur}} = \partial_{\mathrm{cur}}\mathcal{Q}_{\mathrm{cur}}, \qquad \mathcal{C}_{\mathrm{hom}} = \partial_{\mathrm{hom}}\mathcal{Q}_{\mathrm{hom}}. \tag{13}$$

The same note also fixes ambient compatibility sectors

$$\operatorname{ran}\widetilde{\Pi}_{\mathrm{cmp,car}} \subset \mathscr{E}_{\mathrm{car}}, \qquad \operatorname{ran}\widetilde{\Pi}_{\mathrm{cmp,cur}} \subset \mathscr{E}_{\mathrm{cur}}, \qquad \operatorname{ran}\widetilde{\Pi}_{\mathrm{cmp,hom}} \subset \mathscr{E}_{\mathrm{hom}},$$

with exact-preserving ambient equilibrium energies

$$\mathcal{U}_{\mathrm{car}} : \operatorname{ran} \widetilde{\Pi}_{\mathrm{cmp,car}} \to \mathbb{R}, \qquad \mathcal{U}_{\mathrm{cur}} : \operatorname{ran} \widetilde{\Pi}_{\mathrm{cmp,cur}} \to \mathbb{R}, \qquad \mathcal{U}_{\mathrm{hom}} : \operatorname{ran} \widetilde{\Pi}_{\mathrm{cmp,hom}} \to \mathbb{R}$$

and closed C^2 equilibrium manifolds

$$\mathcal{N}_{\mathrm{car}} \subset \operatorname{ran} \widetilde{\Pi}_{\mathrm{cmp,car}}, \qquad \mathcal{N}_{\mathrm{cur}} \subset \operatorname{ran} \widetilde{\Pi}_{\mathrm{cmp,cur}}, \qquad \mathcal{N}_{\mathrm{hom}} \subset \operatorname{ran} \widetilde{\Pi}_{\mathrm{cmp,hom}}$$

through the origin. Their second Fréchet derivatives are represented by bounded self-adjoint ambient Hessians

$$\mathbb{H}_{\mathrm{car}} \equiv D^2\mathcal{U}_{\mathrm{car}}(0), \qquad \mathbb{H}_{\mathrm{cur}} \equiv D^2\mathcal{U}_{\mathrm{cur}}(0), \qquad \mathbb{H}_{\mathrm{hom}} \equiv D^2\mathcal{U}_{\mathrm{hom}}(0), \tag{14}$$

whose orthogonal kernel projectors are the tangent equilibrium projectors

$$\widetilde{\Pi}_{\mathrm{mic,car}}, \qquad \widetilde{\Pi}_{\mathrm{mic,cur}}, \qquad \widetilde{\Pi}_{\mathrm{mic,hom}},$$

and whose strict transverse coercivity yields the positive constants

$$\rho_{\mathrm{eq,car}}, \qquad \rho_{\mathrm{eq,cur}}, \qquad \rho_{\mathrm{eq,hom}}.$$

Pulling them back along the fixed exact-preserving readouts gives the primitive response operators

$$\mathcal{R}_{\mathrm{sel,car}} = \mathcal{A}_{\mathrm{car}}^{*}\mathbb{H}_{\mathrm{car}}\mathcal{A}_{\mathrm{car}}, \qquad \mathcal{R}_{\mathrm{sel,cur}} = \mathcal{A}_{\mathrm{cur}}^{*}\mathbb{H}_{\mathrm{cur}}\mathcal{A}_{\mathrm{cur}}, \qquad \mathcal{R}_{\mathrm{sel,hom}} = \mathcal{A}_{\mathrm{hom}}^{*}\mathbb{H}_{\mathrm{hom}}\mathcal{A}_{\mathrm{hom}}. \tag{15}$$

The recorded microphysical sectors remain

$$\mathscr{Z}_{\mathrm{car}} = \operatorname{ran} \Pi_{\mathrm{sym,car}} \cap \ker \mathcal{C}_{\mathrm{car}}, \qquad \mathscr{Z}_{\mathrm{cur}} = \operatorname{ran} \Pi_{\mathrm{sym,cur}} \cap \ker \mathcal{C}_{\mathrm{cur}}, \qquad \mathscr{Z}_{\mathrm{hom}} = \operatorname{ran} \Pi_{\mathrm{sym,hom}} \cap \ker \mathcal{C}_{\mathrm{hom}}, \tag{16}$$

and the reduced equilibrium Hessians are

$$\begin{aligned} \mathcal{H}_{\mathrm{car}} &\equiv \Pi_{\mathrm{cmp,car}}\mathcal{R}_{\mathrm{sel,car}}\Pi_{\mathrm{cmp,car}}\big|_{\mathscr{Z}_{\mathrm{car}}}, \\ \mathcal{H}_{\mathrm{cur}} &\equiv \Pi_{\mathrm{cmp,cur}}\mathcal{R}_{\mathrm{sel,cur}}\Pi_{\mathrm{cmp,cur}}\big|_{\mathscr{Z}_{\mathrm{cur}}}, \\ \mathcal{H}_{\mathrm{hom}} &\equiv \Pi_{\mathrm{cmp,hom}}\mathcal{R}_{\mathrm{sel,hom}}\Pi_{\mathrm{cmp,hom}}\big|_{\mathscr{Z}_{\mathrm{hom}}}, \end{aligned} \tag{17}$$

with the same microphysical kernel projectors

$$\Pi_{\mathrm{mic,car}}, \qquad \Pi_{\mathrm{mic,cur}}, \qquad \Pi_{\mathrm{mic,hom}}$$

and the same coercive positive gaps

$$\delta_{\mathrm{mic,car}}, \qquad \delta_{\mathrm{mic,cur}}, \qquad \delta_{\mathrm{mic,hom}}.$$

Definition 3 (Recorded symmetry-charge / compatibility-response functional)**.** *The resulting*

recorded symmetry-charge / compatibility-response functional is

$$\begin{aligned}\mathscr{F}_{\mathrm{sel}}[\mathfrak{U},\mathfrak{V},\mathfrak{Q}] \equiv\ & \mathscr{F}_{\mathrm{proto}}[\Pi_{\mathrm{mic,car}}\Pi_{\mathrm{cmp,car}}\Pi_{\mathrm{sym,car}}\mathfrak{U},\ \Pi_{\mathrm{mic,cur}}\Pi_{\mathrm{cmp,cur}}\Pi_{\mathrm{sym,cur}}\mathfrak{V},\ \Pi_{\mathrm{mic,hom}}\Pi_{\mathrm{cmp,hom}}\Pi_{\mathrm{sym,hom}}\mathfrak{Q}] \\ & + \frac{1}{2}\int_{G_{\mathrm{car}}} \|\mathfrak{U} - \mathscr{U}_{\mathrm{car}}(g)\mathfrak{U}\|^2_{\mathrm{root,car}}\, d\mu_{\mathrm{car}}(g) \\ & + \frac{1}{2}\int_{G_{\mathrm{cur}}} \|\mathfrak{V} - \mathscr{U}_{\mathrm{cur}}(g)\mathfrak{V}\|^2_{\mathrm{root,cur}}\, d\mu_{\mathrm{cur}}(g) \\ & + \frac{1}{2}\int_{G_{\mathrm{hom}}} \|\mathfrak{Q} - \mathscr{U}_{\mathrm{hom}}(g)\mathfrak{Q}\|^2_{\mathrm{root,hom}}\, d\mu_{\mathrm{hom}}(g) \\ & + \frac{1}{2}\|\mathcal{C}_{\mathrm{car}}\Pi_{\mathrm{sym,car}}\mathfrak{U}\|^2_{\mathrm{cmp,car}} + \frac{1}{2}\|\mathcal{C}_{\mathrm{cur}}\Pi_{\mathrm{sym,cur}}\mathfrak{V}\|^2_{\mathrm{cmp,cur}} + \frac{1}{2}\|\mathcal{C}_{\mathrm{hom}}\Pi_{\mathrm{sym,hom}}\mathfrak{Q}\|^2_{\mathrm{cmp,hom}} \\ & + \frac{1}{2}\langle \mathcal{R}_{\mathrm{sel,car}}\Pi_{\mathrm{cmp,car}}\Pi_{\mathrm{sym,car}}\mathfrak{U},\ \Pi_{\mathrm{cmp,car}}\Pi_{\mathrm{sym,car}}\mathfrak{U}\rangle_{\mathrm{root,car}} \\ & + \frac{1}{2}\langle \mathcal{R}_{\mathrm{sel,cur}}\Pi_{\mathrm{cmp,cur}}\Pi_{\mathrm{sym,cur}}\mathfrak{V},\ \Pi_{\mathrm{cmp,cur}}\Pi_{\mathrm{sym,cur}}\mathfrak{V}\rangle_{\mathrm{root,cur}} \\ & + \frac{1}{2}\langle \mathcal{R}_{\mathrm{sel,hom}}\Pi_{\mathrm{cmp,hom}}\Pi_{\mathrm{sym,hom}}\mathfrak{Q},\ \Pi_{\mathrm{cmp,hom}}\Pi_{\mathrm{sym,hom}}\mathfrak{Q}\rangle_{\mathrm{root,hom}}.\end{aligned} \tag{18}$$

The current task is to derive the layer entering (18) *while preserving the already fixed chain*

$$\mathscr{F}_{\mathrm{sel}} \Longrightarrow \mathscr{F}_{\mathrm{hess}} = \mathscr{F}_{\mathrm{micro}} \Longrightarrow \mathscr{F}_{\mathrm{deep}} \Longrightarrow \mathscr{F}_{\mathrm{car}} \Longrightarrow \mathscr{F}_{\mathrm{prim}} \Longrightarrow \mathscr{F}_{\mathrm{mic}}. \tag{19}$$

3. Reconfiguration-Group Origin of the Exact-Preserving Derivation Algebras

Definition 4 (Primitive exact-preserving reconfiguration data)**.** *For each sharper sector, let*

$$\mathcal{G}^{\mathrm{rec}}_{\mathrm{car}}, \qquad \mathcal{G}^{\mathrm{rec}}_{\mathrm{cur}}, \qquad \mathcal{G}^{\mathrm{rec}}_{\mathrm{hom}}$$

be finite-dimensional Lie groups with identity elements

$$e_{\mathrm{rec,car}}, \qquad e_{\mathrm{rec,cur}}, \qquad e_{\mathrm{rec,hom}},$$

carrying bi-invariant Riemannian metrics. Let

$$\mathscr{V}_{\mathrm{car}} : \mathcal{G}^{\mathrm{rec}}_{\mathrm{car}} \to \mathcal{U}(\mathscr{U}_{\mathrm{car}}), \qquad \mathscr{V}_{\mathrm{cur}} : \mathcal{G}^{\mathrm{rec}}_{\mathrm{cur}} \to \mathcal{U}(\mathscr{U}_{\mathrm{cur}}), \qquad \mathscr{V}_{\mathrm{hom}} : \mathcal{G}^{\mathrm{rec}}_{\mathrm{hom}} \to \mathcal{U}(\mathscr{U}_{\mathrm{hom}})$$

be strongly continuous exact-preserving unitary actions. Assume:

1. *each action preserves the raw transport data of Definition 5 and the raw equilibrium-reduction data of Definition 6;*
2. *the exact-preserving branch corresponds to the identity configurations;*
3. *the infinitesimal action kernels*

$$\mathfrak{k}_{\mathrm{rec,car}} \equiv \left\{ X \in T_{e_{\mathrm{rec,car}}}\mathcal{G}^{\mathrm{rec}}_{\mathrm{car}} : \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathscr{V}_{\mathrm{car}}(\exp(tX))\mathfrak{U} = 0 \textit{ for all } \mathfrak{U} \in \mathscr{U}_{\mathrm{car}} \right\}, \tag{20}$$

$$\mathfrak{k}_{\mathrm{rec,cur}} \equiv \left\{ Y \in T_{e_{\mathrm{rec,cur}}}\mathcal{G}^{\mathrm{rec}}_{\mathrm{cur}} : \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathscr{V}_{\mathrm{cur}}(\exp(tY))\mathfrak{V} = 0 \textit{ for all } \mathfrak{V} \in \mathscr{U}_{\mathrm{cur}} \right\}, \tag{21}$$

$$\mathfrak{k}_{\mathrm{rec,hom}} \equiv \left\{ Z \in T_{e_{\mathrm{rec,hom}}}\mathcal{G}^{\mathrm{rec}}_{\mathrm{hom}} : \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathscr{V}_{\mathrm{hom}}(\exp(tZ))\mathfrak{Q} = 0 \textit{ for all } \mathfrak{Q} \in \mathscr{U}_{\mathrm{hom}} \right\} \tag{22}$$

are ideals in the corresponding tangent Lie algebras.

Define the visible tangent quotients

$$\mathfrak{d}_{\mathrm{car}} \equiv T_{e_{\mathrm{rec,car}}}\mathcal{G}^{\mathrm{rec}}_{\mathrm{car}}/\mathfrak{k}_{\mathrm{rec,car}}, \qquad \mathfrak{d}_{\mathrm{cur}} \equiv T_{e_{\mathrm{rec,cur}}}\mathcal{G}^{\mathrm{rec}}_{\mathrm{cur}}/\mathfrak{k}_{\mathrm{rec,cur}}, \qquad \mathfrak{d}_{\mathrm{hom}} \equiv T_{e_{\mathrm{rec,hom}}}\mathcal{G}^{\mathrm{rec}}_{\mathrm{hom}}/\mathfrak{k}_{\mathrm{rec,hom}}, \tag{23}$$

and identify them with their induced infinitesimal actions

$$[X]\mathfrak{U} \equiv \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathscr{V}_{\mathrm{car}}(\exp(tX))\mathfrak{U}, \tag{24}$$

$$[Y]\mathfrak{V} \equiv \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathscr{V}_{\mathrm{cur}}(\exp(tY))\mathfrak{V}, \tag{25}$$

$$[Z]\mathfrak{Q} \equiv \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathscr{V}_{\mathrm{hom}}(\exp(tZ))\mathfrak{Q}. \tag{26}$$

Proposition 1 (The exact-preserving derivation algebras are forced by reconfiguration groups)**.** *The reconfiguration data of Definition 4 recover exactly the recorded derivation-algebra layer.*

1. *The quotients in* (23) *are finite-dimensional real Lie algebras.*
2. *Their induced actions* (24)–(26) *are bounded skew-adjoint exact-preserving derivations on the sharper sectors.*
3. *The bi-invariant metrics on the reconfiguration groups descend to positive-definite* Ad*-invariant inner products on*
$$\mathfrak{d}_{\mathrm{car}}, \qquad \mathfrak{d}_{\mathrm{cur}}, \qquad \mathfrak{d}_{\mathrm{hom}}.$$
4. *Consequently the derivation algebras of Definition 2 are no longer primitive at present scope; they are exactly the visible tangent quotients of the deeper exact-preserving reconfiguration groups.*

Proof. The tangent spaces at the identity of finite-dimensional Lie groups are finite-dimensional Lie algebras with bracket induced by commutators of left-invariant vector fields. Because the infinitesimal action kernels $\mathfrak{k}_{\mathrm{rec,car}}, \mathfrak{k}_{\mathrm{rec,cur}}, \mathfrak{k}_{\mathrm{rec,hom}}$ are ideals, the quotients in (23) are well-defined finite-dimensional Lie algebras.

Differentiating a strongly continuous unitary action at the identity yields skew-adjoint infinitesimal generators on the corresponding Hilbert sectors. Because the actions are exact preserving and leave the raw transport and raw equilibrium-reduction data invariant, their infinitesimal generators preserve those deeper structures as well. Pulling back along the visible embeddings and reduction maps of Definitions 5 and 6, one obtains exactly the recorded notion of exact-preserving derivation on the sharper sectors.

For a Lie group with bi-invariant Riemannian metric, the metric at the identity is Ad-invariant on the tangent Lie algebra. Since the infinitesimal action kernels are orthogonal ideals, the metric descends to a positive-definite Ad-invariant inner product on each quotient. The induced quotient actions are faithful by construction. Hence the visible tangent quotients are exactly the recorded derivation algebras. □

4. Transport-Law Origin of the Primitive Density / Balance Layer

Definition 5 (Primitive substrate transport data)**.** *On the symmetry-fixed sectors*

$$\operatorname{ran}\Pi_{\mathrm{sym,car}}, \qquad \operatorname{ran}\Pi_{\mathrm{sym,cur}}, \qquad \operatorname{ran}\Pi_{\mathrm{sym,hom}},$$

let

$$\mathcal{S}_{\mathrm{car}}, \qquad \mathcal{S}_{\mathrm{cur}}, \qquad \mathcal{S}_{\mathrm{hom}}$$

be Hilbert manifolds with exact-preserving base states

$$s^0_{\mathrm{car}}, \qquad s^0_{\mathrm{cur}}, \qquad s^0_{\mathrm{hom}}.$$

Let

$$\iota_{\mathrm{car}} : \operatorname{ran}\Pi_{\mathrm{sym,car}} \to \mathcal{S}_{\mathrm{car}}, \qquad \iota_{\mathrm{cur}} : \operatorname{ran}\Pi_{\mathrm{sym,cur}} \to \mathcal{S}_{\mathrm{cur}}, \qquad \iota_{\mathrm{hom}} : \operatorname{ran}\Pi_{\mathrm{sym,hom}} \to \mathcal{S}_{\mathrm{hom}}$$

be C^2 exact-preserving embeddings satisfying

$$\iota_{\mathrm{car}}(0) = s^0_{\mathrm{car}}, \qquad \iota_{\mathrm{cur}}(0) = s^0_{\mathrm{cur}}, \qquad \iota_{\mathrm{hom}}(0) = s^0_{\mathrm{hom}}, \tag{27}$$

and assume their linearizations at the origin are isometric injections.

Let

$$\boldsymbol{\mathcal{Q}}_{\mathrm{car}} : \mathcal{S}_{\mathrm{car}} \to \mathscr{N}_{\mathrm{car}}, \qquad \boldsymbol{\mathcal{Q}}_{\mathrm{cur}} : \mathcal{S}_{\mathrm{cur}} \to \mathscr{N}_{\mathrm{cur}}, \qquad \boldsymbol{\mathcal{Q}}_{\mathrm{hom}} : \mathcal{S}_{\mathrm{hom}} \to \mathscr{N}_{\mathrm{hom}}$$

be C^2 raw density observables,

$$\mathbb{V}_{\mathrm{car}} \in \Gamma(T\mathcal{S}_{\mathrm{car}}), \qquad \mathbb{V}_{\mathrm{cur}} \in \Gamma(T\mathcal{S}_{\mathrm{cur}}), \qquad \mathbb{V}_{\mathrm{hom}} \in \Gamma(T\mathcal{S}_{\mathrm{hom}})$$

be C^1 transport vector fields, and

$$\mathbb{J}_{\mathrm{car}} : \mathcal{S}_{\mathrm{car}} \to \mathscr{J}_{\mathrm{car}}, \qquad \mathbb{J}_{\mathrm{cur}} : \mathcal{S}_{\mathrm{cur}} \to \mathscr{J}_{\mathrm{cur}}, \qquad \mathbb{J}_{\mathrm{hom}} : \mathcal{S}_{\mathrm{hom}} \to \mathscr{J}_{\mathrm{hom}}$$

be C^1 raw flux maps. Let

$$\mathbb{B}_{\mathrm{car}} : \mathscr{J}_{\mathrm{car}} \to \mathscr{M}_{\mathrm{car}}, \qquad \mathbb{B}_{\mathrm{cur}} : \mathscr{J}_{\mathrm{cur}} \to \mathscr{M}_{\mathrm{cur}}, \qquad \mathbb{B}_{\mathrm{hom}} : \mathscr{J}_{\mathrm{hom}} \to \mathscr{M}_{\mathrm{hom}}$$

be bounded linear flux-divergence operators. Assume the exact-preserving base states are stationary and unsourced:

$$\boldsymbol{\mathcal{Q}}_{\mathrm{car}}(s^0_{\mathrm{car}}) = 0, \qquad \boldsymbol{\mathcal{Q}}_{\mathrm{cur}}(s^0_{\mathrm{cur}}) = 0, \qquad \boldsymbol{\mathcal{Q}}_{\mathrm{hom}}(s^0_{\mathrm{hom}}) = 0, \tag{28}$$

$$\mathbb{V}_{\mathrm{car}}(s^0_{\mathrm{car}}) = 0, \qquad \mathbb{V}_{\mathrm{cur}}(s^0_{\mathrm{cur}}) = 0, \qquad \mathbb{V}_{\mathrm{hom}}(s^0_{\mathrm{hom}}) = 0, \tag{29}$$

$$\mathbb{J}_{\mathrm{car}}(s^0_{\mathrm{car}}) = 0, \qquad \mathbb{J}_{\mathrm{cur}}(s^0_{\mathrm{cur}}) = 0, \qquad \mathbb{J}_{\mathrm{hom}}(s^0_{\mathrm{hom}}) = 0. \tag{30}$$

Define the raw balance defects by

$$\mathfrak{c}_{\mathrm{car}}(s) \equiv D\boldsymbol{\mathcal{Q}}_{\mathrm{car}}(s)[\mathbb{V}_{\mathrm{car}}(s)] + \mathbb{B}_{\mathrm{car}}\mathbb{J}_{\mathrm{car}}(s), \tag{31}$$

$$\mathfrak{c}_{\mathrm{cur}}(r) \equiv D\boldsymbol{\mathcal{Q}}_{\mathrm{cur}}(r)[\mathbb{V}_{\mathrm{cur}}(r)] + \mathbb{B}_{\mathrm{cur}}\mathbb{J}_{\mathrm{cur}}(r), \tag{32}$$

$$\mathfrak{c}_{\mathrm{hom}}(h) \equiv D\boldsymbol{\mathcal{Q}}_{\mathrm{hom}}(h)[\mathbb{V}_{\mathrm{hom}}(h)] + \mathbb{B}_{\mathrm{hom}}\mathbb{J}_{\mathrm{hom}}(h). \tag{33}$$

The recorded primitive nonlinear density and balance laws are then defined by pullback:

$$\widehat{\mathcal{Q}}_{\mathrm{car}} \equiv \boldsymbol{\mathcal{Q}}_{\mathrm{car}} \circ \iota_{\mathrm{car}}, \qquad \widehat{\mathcal{Q}}_{\mathrm{cur}} \equiv \boldsymbol{\mathcal{Q}}_{\mathrm{cur}} \circ \iota_{\mathrm{cur}}, \qquad \widehat{\mathcal{Q}}_{\mathrm{hom}} \equiv \boldsymbol{\mathcal{Q}}_{\mathrm{hom}} \circ \iota_{\mathrm{hom}}, \tag{34}$$

$$\widehat{\mathfrak{c}}_{\mathrm{car}} \equiv \mathfrak{c}_{\mathrm{car}} \circ \iota_{\mathrm{car}}, \qquad \widehat{\mathfrak{c}}_{\mathrm{cur}} \equiv \mathfrak{c}_{\mathrm{cur}} \circ \iota_{\mathrm{cur}}, \qquad \widehat{\mathfrak{c}}_{\mathrm{hom}} \equiv \mathfrak{c}_{\mathrm{hom}} \circ \iota_{\mathrm{hom}}. \tag{35}$$

Define the recorded density readouts by linearization:

$$\mathcal{Q}_{\mathrm{car}} \equiv D\widehat{\mathcal{Q}}_{\mathrm{car}}(0), \qquad \mathcal{Q}_{\mathrm{cur}} \equiv D\widehat{\mathcal{Q}}_{\mathrm{cur}}(0), \qquad \mathcal{Q}_{\mathrm{hom}} \equiv D\widehat{\mathcal{Q}}_{\mathrm{hom}}(0). \tag{36}$$

Assume the visible density ranges are closed and that

$$\ker \mathcal{Q}_{\mathrm{car}} \subseteq \ker D\widehat{\mathfrak{c}}_{\mathrm{car}}(0), \qquad \ker \mathcal{Q}_{\mathrm{cur}} \subseteq \ker D\widehat{\mathfrak{c}}_{\mathrm{cur}}(0), \qquad \ker \mathcal{Q}_{\mathrm{hom}} \subseteq \ker D\widehat{\mathfrak{c}}_{\mathrm{hom}}(0). \tag{37}$$

Then there exist unique bounded operators on the visible density ranges,

$$\partial_{\mathrm{car}} : \operatorname{ran} \mathcal{Q}_{\mathrm{car}} \to \mathscr{M}_{\mathrm{car}}, \qquad \partial_{\mathrm{cur}} : \operatorname{ran} \mathcal{Q}_{\mathrm{cur}} \to \mathscr{M}_{\mathrm{cur}}, \qquad \partial_{\mathrm{hom}} : \operatorname{ran} \mathcal{Q}_{\mathrm{hom}} \to \mathscr{M}_{\mathrm{hom}}, \tag{38}$$

satisfying

$$D\widehat{\mathfrak{c}}_{\mathrm{car}}(0) = \partial_{\mathrm{car}}\mathcal{Q}_{\mathrm{car}}, \qquad D\widehat{\mathfrak{c}}_{\mathrm{cur}}(0) = \partial_{\mathrm{cur}}\mathcal{Q}_{\mathrm{cur}}, \qquad D\widehat{\mathfrak{c}}_{\mathrm{hom}}(0) = \partial_{\mathrm{hom}}\mathcal{Q}_{\mathrm{hom}}. \tag{39}$$

As in the recorded layer, extend them by zero on the orthogonal complements of the visible density ranges inside $\mathscr{N}_{\mathrm{car}}, \mathscr{N}_{\mathrm{cur}}, \mathscr{N}_{\mathrm{hom}}$, and identify the compatibility-current maps with the balance defects:

$$\mathfrak{c}_{\mathrm{car}} \equiv \widehat{\mathfrak{c}}_{\mathrm{car}}, \qquad \mathfrak{c}_{\mathrm{cur}} \equiv \widehat{\mathfrak{c}}_{\mathrm{cur}}, \qquad \mathfrak{c}_{\mathrm{hom}} \equiv \widehat{\mathfrak{c}}_{\mathrm{hom}}. \tag{40}$$

Proposition 2 (The primitive density / balance layer is forced by transport laws)**.** *At current quadratic exact-preservation scope, the transport data of Definition 5 recover exactly the recorded primitive density / balance layer.*

1. *The recorded nonlinear density maps and balance defects are the pullbacks* (34) *and* (35) *of raw substrate transport laws.*

2. *Their linearizations are exactly the recorded density readouts*

$$\mathcal{Q}_{\mathrm{car}}, \qquad \mathcal{Q}_{\mathrm{cur}}, \qquad \mathcal{Q}_{\mathrm{hom}}$$

   and the recorded continuity operators

$$\partial_{\mathrm{car}}, \qquad \partial_{\mathrm{cur}}, \qquad \partial_{\mathrm{hom}}.$$

3. *The recorded compatibility-current maps are the same balance defects:*

$$\mathfrak{c}_{\mathrm{car}} = \widehat{\mathfrak{c}}_{\mathrm{car}}, \qquad \mathfrak{c}_{\mathrm{cur}} = \widehat{\mathfrak{c}}_{\mathrm{cur}}, \qquad \mathfrak{c}_{\mathrm{hom}} = \widehat{\mathfrak{c}}_{\mathrm{hom}},$$

and their linearizations are exactly

$$D\mathfrak{c}_{\mathrm{car}}(0) = \partial_{\mathrm{car}}\mathcal{Q}_{\mathrm{car}}, \qquad D\mathfrak{c}_{\mathrm{cur}}(0) = \partial_{\mathrm{cur}}\mathcal{Q}_{\mathrm{cur}}, \qquad D\mathfrak{c}_{\mathrm{hom}}(0) = \partial_{\mathrm{hom}}\mathcal{Q}_{\mathrm{hom}}. \tag{41}$$

4. Therefore the recorded compatibility operators are again

$$\mathcal{C}_{\mathrm{car}} = \partial_{\mathrm{car}}\mathcal{Q}_{\mathrm{car}}, \qquad \mathcal{C}_{\mathrm{cur}} = \partial_{\mathrm{cur}}\mathcal{Q}_{\mathrm{cur}}, \qquad \mathcal{C}_{\mathrm{hom}} = \partial_{\mathrm{hom}}\mathcal{Q}_{\mathrm{hom}}. \tag{42}$$

Hence the primitive nonlinear density and balance laws are no longer primitive at present scope; they are the visible pullbacks of a still more primitive substrate transport layer.

Proof. Item (1) is the defining pullback construction (34) and (35). The vanishing conditions (28)–(30) imply

$$\widehat{\mathcal{Q}}_{\mathrm{car}}(0) = 0, \qquad \widehat{\mathcal{Q}}_{\mathrm{cur}}(0) = 0, \qquad \widehat{\mathcal{Q}}_{\mathrm{hom}}(0) = 0,$$

and likewise

$$\widehat{\mathfrak{c}}_{\mathrm{car}}(0) = 0, \qquad \widehat{\mathfrak{c}}_{\mathrm{cur}}(0) = 0, \qquad \widehat{\mathfrak{c}}_{\mathrm{hom}}(0) = 0.$$

Item (2) follows by differentiating the pullback definitions at the exact-preserving branch. The factorization assumption (37) then guarantees unique bounded operators on the visible density ranges satisfying (39); this is the same quotient-by-kernel argument used in the recorded layer.

Item (3) is the definition (40) together with (39). Finally (42) is exactly the recorded identification of the compatibility operators with the linearized balance defects. □

5. Equilibrium-Reduction Origin of the Primitive Equilibrium Layer

Definition 6 (Primitive raw equilibrium-reduction data)**.** *On the ambient compatibility sectors*

$$\operatorname{ran}\widetilde{\Pi}_{\mathrm{cmp,car}} \subset \mathscr{E}_{\mathrm{car}}, \qquad \operatorname{ran}\widetilde{\Pi}_{\mathrm{cmp,cur}} \subset \mathscr{E}_{\mathrm{cur}}, \qquad \operatorname{ran}\widetilde{\Pi}_{\mathrm{cmp,hom}} \subset \mathscr{E}_{\mathrm{hom}},$$

let

$$\widetilde{\mathscr{E}}_{\mathrm{car}}, \qquad \widetilde{\mathscr{E}}_{\mathrm{cur}}, \qquad \widetilde{\mathscr{E}}_{\mathrm{hom}}$$

be Hilbert manifolds with distinguished origins 0*. Let*

$$\mathfrak{R}_{\mathrm{car}} : \operatorname{ran}\widetilde{\Pi}_{\mathrm{cmp,car}} \to \widetilde{\mathscr{E}}_{\mathrm{car}}, \qquad \mathfrak{R}_{\mathrm{cur}} : \operatorname{ran}\widetilde{\Pi}_{\mathrm{cmp,cur}} \to \widetilde{\mathscr{E}}_{\mathrm{cur}}, \qquad \mathfrak{R}_{\mathrm{hom}} : \operatorname{ran}\widetilde{\Pi}_{\mathrm{cmp,hom}} \to \widetilde{\mathscr{E}}_{\mathrm{hom}}$$

be C^2 exact-preserving reduction maps satisfying

$$\mathfrak{R}_{\mathrm{car}}(0) = 0, \qquad \mathfrak{R}_{\mathrm{cur}}(0) = 0, \qquad \mathfrak{R}_{\mathrm{hom}}(0) = 0, \tag{43}$$

and assume their linearizations at the origin are isometric injections.

Let

$$\widetilde{\mathcal{U}}_{\mathrm{car}} : \widetilde{\mathscr{E}}_{\mathrm{car}} \to \mathbb{R}, \qquad \widetilde{\mathcal{U}}_{\mathrm{cur}} : \widetilde{\mathscr{E}}_{\mathrm{cur}} \to \mathbb{R}, \qquad \widetilde{\mathcal{U}}_{\mathrm{hom}} : \widetilde{\mathscr{E}}_{\mathrm{hom}} \to \mathbb{R}$$

be C^2 exact-preserving raw equilibrium energies, and let

$$\widetilde{\mathcal{N}}_{\mathrm{car}} \subset \widetilde{\mathscr{E}}_{\mathrm{car}}, \qquad \widetilde{\mathcal{N}}_{\mathrm{cur}} \subset \widetilde{\mathscr{E}}_{\mathrm{cur}}, \qquad \widetilde{\mathcal{N}}_{\mathrm{hom}} \subset \widetilde{\mathscr{E}}_{\mathrm{hom}}$$

be closed C^2 equilibrium manifolds through the origin. Assume:

1. *each raw energy is constant on its raw equilibrium manifold and has vanishing first derivative there;*

2. *the raw second Fréchet derivatives at the origin are represented by bounded self-adjoint operators;*

3. *strict raw transverse coercivity holds near the origin:*

$$\widetilde{\mathcal{U}}_{\mathrm{car}}(\eta) \geq \frac{\widetilde{\rho}_{\mathrm{eq,car}}}{2}\,\mathrm{dist}(\eta, \widetilde{\mathcal{N}}_{\mathrm{car}})^2 + o(\|\eta\|^2_{\mathrm{raw,car}}), \qquad \widetilde{\rho}_{\mathrm{eq,car}} > 0, \tag{44}$$

$$\widetilde{\mathcal{U}}_{\mathrm{cur}}(\zeta) \geq \frac{\widetilde{\rho}_{\mathrm{eq,cur}}}{2}\,\mathrm{dist}(\zeta, \widetilde{\mathcal{N}}_{\mathrm{cur}})^2 + o(\|\zeta\|^2_{\mathrm{raw,cur}}), \qquad \widetilde{\rho}_{\mathrm{eq,cur}} > 0, \tag{45}$$

$$\widetilde{\mathcal{U}}_{\mathrm{hom}}(\omega) \geq \frac{\widetilde{\rho}_{\mathrm{eq,hom}}}{2}\,\mathrm{dist}(\omega, \widetilde{\mathcal{N}}_{\mathrm{hom}})^2 + o(\|\omega\|^2_{\mathrm{raw,hom}}), \qquad \widetilde{\rho}_{\mathrm{eq,hom}} > 0. \tag{46}$$

Define the recorded primitive ambient equilibrium energies and equilibrium manifolds by pullback:

$$\mathcal{U}_{\mathrm{car}} \equiv \widetilde{\mathcal{U}}_{\mathrm{car}} \circ \mathfrak{R}_{\mathrm{car}}, \qquad \mathcal{U}_{\mathrm{cur}} \equiv \widetilde{\mathcal{U}}_{\mathrm{cur}} \circ \mathfrak{R}_{\mathrm{cur}}, \qquad \mathcal{U}_{\mathrm{hom}} \equiv \widetilde{\mathcal{U}}_{\mathrm{hom}} \circ \mathfrak{R}_{\mathrm{hom}}, \tag{47}$$

$$\mathcal{N}_{\mathrm{car}} \equiv \mathfrak{R}^{-1}_{\mathrm{car}}(\widetilde{\mathcal{N}}_{\mathrm{car}}), \qquad \mathcal{N}_{\mathrm{cur}} \equiv \mathfrak{R}^{-1}_{\mathrm{cur}}(\widetilde{\mathcal{N}}_{\mathrm{cur}}), \qquad \mathcal{N}_{\mathrm{hom}} \equiv \mathfrak{R}^{-1}_{\mathrm{hom}}(\widetilde{\mathcal{N}}_{\mathrm{hom}}). \tag{48}$$

Because the reduction maps are C^2 local embeddings with isometric linearizations, the raw coercive gaps induce visible positive constants

$$\rho_{\mathrm{eq,car}}, \qquad \rho_{\mathrm{eq,cur}}, \qquad \rho_{\mathrm{eq,hom}}$$

for which

$$\mathcal{U}_{\mathrm{car}}(\eta) \geq \frac{\rho_{\mathrm{eq,car}}}{2}\,\mathrm{dist}(\eta, \mathcal{N}_{\mathrm{car}})^2 + o(\|\eta\|^2_{\mathrm{eq,car}}), \tag{49}$$

$$\mathcal{U}_{\mathrm{cur}}(\zeta) \geq \frac{\rho_{\mathrm{eq,cur}}}{2}\,\mathrm{dist}(\zeta, \mathcal{N}_{\mathrm{cur}})^2 + o(\|\zeta\|^2_{\mathrm{eq,cur}}), \tag{50}$$

$$\mathcal{U}_{\mathrm{hom}}(\omega) \geq \frac{\rho_{\mathrm{eq,hom}}}{2}\,\mathrm{dist}(\omega, \mathcal{N}_{\mathrm{hom}})^2 + o(\|\omega\|^2_{\mathrm{eq,hom}}). \tag{51}$$

Let $\widetilde{\Pi}_{\mathrm{mic,car}}, \widetilde{\Pi}_{\mathrm{mic,cur}}, \widetilde{\Pi}_{\mathrm{mic,hom}}$ denote the orthogonal projectors onto the tangent spaces

$$T_0\mathcal{N}_{\mathrm{car}}, \qquad T_0\mathcal{N}_{\mathrm{cur}}, \qquad T_0\mathcal{N}_{\mathrm{hom}}.$$

Proposition 3 (The primitive equilibrium layer is forced by equilibrium reduction)**.** *The reduction data of Definition 6 recover exactly the recorded primitive equilibrium layer.*

1. *The recorded primitive ambient equilibrium energies and equilibrium manifolds are exactly the pullbacks* (47) *and* (48) *of the raw equilibrium geometry.*

2. *Their ambient Hessians are the second variations*

$$\mathbb{H}_{\mathrm{car}} = D\mathfrak{R}_{\mathrm{car}}(0)^* D^2\widetilde{\mathcal{U}}_{\mathrm{car}}(0) D\mathfrak{R}_{\mathrm{car}}(0), \tag{52}$$

$$\mathbb{H}_{\mathrm{cur}} = D\mathfrak{R}_{\mathrm{cur}}(0)^* D^2\widetilde{\mathcal{U}}_{\mathrm{cur}}(0) D\mathfrak{R}_{\mathrm{cur}}(0), \tag{53}$$

$$\mathbb{H}_{\mathrm{hom}} = D\mathfrak{R}_{\mathrm{hom}}(0)^* D^2\widetilde{\mathcal{U}}_{\mathrm{hom}}(0) D\mathfrak{R}_{\mathrm{hom}}(0), \tag{54}$$

hence they are bounded, self adjoint, and nonnegative.

3. *Their kernels are exactly the tangent equilibrium directions:*

$$\ker \mathbb{H}_{\mathrm{car}} = T_0\mathcal{N}_{\mathrm{car}}, \qquad \ker \mathbb{H}_{\mathrm{cur}} = T_0\mathcal{N}_{\mathrm{cur}}, \qquad \ker \mathbb{H}_{\mathrm{hom}} = T_0\mathcal{N}_{\mathrm{hom}}. \tag{55}$$

Therefore $\widetilde{\Pi}_{\mathrm{mic,car}}, \widetilde{\Pi}_{\mathrm{mic,cur}}, \widetilde{\Pi}_{\mathrm{mic,hom}}$ *are exactly the orthogonal kernel projectors.*

4. *The recorded positive-gap estimates hold:*

$$\langle \mathbb{H}_{\mathrm{car}}\eta, \eta\rangle_{\mathrm{eq,car}} \geq \rho_{\mathrm{eq,car}} \left\| (\mathbf{1} - \widetilde{\Pi}_{\mathrm{mic,car}})\eta \right\|^2_{\mathrm{eq,car}}, \tag{56}$$

$$\langle \mathbb{H}_{\mathrm{cur}}\zeta, \zeta\rangle_{\mathrm{eq,cur}} \geq \rho_{\mathrm{eq,cur}} \left\| (\mathbf{1} - \widetilde{\Pi}_{\mathrm{mic,cur}})\zeta \right\|^2_{\mathrm{eq,cur}}, \tag{57}$$

$$\langle \mathbb{H}_{\mathrm{hom}}\omega, \omega\rangle_{\mathrm{eq,hom}} \geq \rho_{\mathrm{eq,hom}} \left\| (\mathbf{1} - \widetilde{\Pi}_{\mathrm{mic,hom}})\omega \right\|^2_{\mathrm{eq,hom}}. \tag{58}$$

5. *Consequently the same primitive response operators*

$$\mathcal{R}_{\mathrm{sel,car}}, \qquad \mathcal{R}_{\mathrm{sel,cur}}, \qquad \mathcal{R}_{\mathrm{sel,hom}}$$

are recovered via (15)*, and therefore the same reduced Hessians*

$$\mathcal{H}_{\mathrm{car}}, \qquad \mathcal{H}_{\mathrm{cur}}, \qquad \mathcal{H}_{\mathrm{hom}}$$

the same microphysical kernel projectors

$$\Pi_{\mathrm{mic,car}}, \qquad \Pi_{\mathrm{mic,cur}}, \qquad \Pi_{\mathrm{mic,hom}}$$

and the same coercive positive gaps

$$\delta_{\mathrm{mic,car}}, \qquad \delta_{\mathrm{mic,cur}}, \qquad \delta_{\mathrm{mic,hom}}$$

are recovered as well.

Thus the primitive equilibrium energies and equilibrium manifolds are no longer primitive at present scope; they are the visible reduced image of a deeper equilibrium-reduction layer.

Proof. Item (1) is the pullback definition itself. Because the reduction maps are C^2, the chain rule gives the Hessian identities (52)–(54), proving item (2). Bounded self-adjointness follows because the raw second variations are bounded symmetric bilinear forms and the reduction linearizations are bounded.

If $v \in T_0\mathcal{N}_{\mathrm{car}}$, then there is a C^2 curve $\gamma(t) \subset \mathcal{N}_{\mathrm{car}}$ with $\gamma(0) = 0$ and $\gamma'(0) = v$. Since $\mathcal{U}_{\mathrm{car}}$ is constant on $\mathcal{N}_{\mathrm{car}}$, differentiating twice gives

$$\langle \mathbb{H}_{\mathrm{car}}v, v\rangle_{\mathrm{eq,car}} = 0,$$

so $T_0\mathcal{N}_{\mathrm{car}} \subseteq \ker \mathbb{H}_{\mathrm{car}}$. Conversely, if $v \perp T_0\mathcal{N}_{\mathrm{car}}$, then applying (49) to $\eta = tv$ and dividing by t^2 yields

$$\frac{1}{2}\langle \mathbb{H}_{\mathrm{car}}v, v\rangle_{\mathrm{eq,car}} \geq \frac{\rho_{\mathrm{eq,car}}}{2}\|v\|^2_{\mathrm{eq,car}}.$$

Hence no nonzero vector orthogonal to $T_0\mathcal{N}_{\mathrm{car}}$ can lie in the kernel. The relabeling and homogeneous

sectors are identical, proving (55) and the gap estimates (56)–(58).

Finally, once the same ambient Hessians, ambient kernel projectors, and visible gap constants are recovered, the fixed exact-preserving readouts $\mathcal{A}_{\mathrm{car}}, \mathcal{A}_{\mathrm{cur}}, \mathcal{A}_{\mathrm{hom}}$ recover the same primitive response operators by (15). Because the symmetry projectors, compatibility operators, and microphysical sectors are unchanged, the same reduced Hessians, microphysical kernel projectors, and coercive positive gaps are recovered verbatim. □

6. Exact Recovery of the Recorded Charge-Generator / Continuity-Hessian Layer

Theorem 1 (Same derivation-law / balance-law / equilibrium-manifold layer)**.** *The deeper reconfiguration-group / transport-law / equilibrium-reduction construction introduced above recovers exactly the recorded derivation-law / balance-law / equilibrium-manifold layer:*

1. *Proposition 1 recovers the same exact-preserving derivation algebras* $\mathfrak{d}_{\mathrm{car}}, \mathfrak{d}_{\mathrm{cur}}, \mathfrak{d}_{\mathrm{hom}}$*. Therefore the same visible charge Lie algebras* $\mathfrak{g}_{\mathrm{car}}, \mathfrak{g}_{\mathrm{cur}}, \mathfrak{g}_{\mathrm{hom}}$*, the same generator maps* $\Gamma_{\mathrm{car}}, \Gamma_{\mathrm{cur}}, \Gamma_{\mathrm{hom}}$*, the same compact symmetry actions* $\mathscr{U}_{\mathrm{car}}, \mathscr{U}_{\mathrm{cur}}, \mathscr{U}_{\mathrm{hom}}$*, the same Casimir operators* $\mathbb{C}_{\mathrm{car}}, \mathbb{C}_{\mathrm{cur}}, \mathbb{C}_{\mathrm{hom}}$*, and the same symmetry projectors* $\Pi_{\mathrm{sym,car}}, \Pi_{\mathrm{sym,cur}}, \Pi_{\mathrm{sym,hom}}$ *follow again.*

2. *Proposition 2 recovers the same primitive nonlinear density maps* $\widehat{\mathcal{Q}}_{\mathrm{car}}, \widehat{\mathcal{Q}}_{\mathrm{cur}}, \widehat{\mathcal{Q}}_{\mathrm{hom}}$*, the same balance defects* $\widehat{\mathfrak{c}}_{\mathrm{car}}, \widehat{\mathfrak{c}}_{\mathrm{cur}}, \widehat{\mathfrak{c}}_{\mathrm{hom}}$*, the same linear density readouts* $\mathcal{Q}_{\mathrm{car}}, \mathcal{Q}_{\mathrm{cur}}, \mathcal{Q}_{\mathrm{hom}}$*, the same continuity operators* $\partial_{\mathrm{car}}, \partial_{\mathrm{cur}}, \partial_{\mathrm{hom}}$*, the same compatibility-current maps* $\mathfrak{c}_{\mathrm{car}}, \mathfrak{c}_{\mathrm{cur}}, \mathfrak{c}_{\mathrm{hom}}$*, and the same compatibility operators* $\mathcal{C}_{\mathrm{car}}, \mathcal{C}_{\mathrm{cur}}, \mathcal{C}_{\mathrm{hom}}$*.*

3. *Proposition 3 recovers the same primitive ambient equilibrium energies* $\mathcal{U}_{\mathrm{car}}, \mathcal{U}_{\mathrm{cur}}, \mathcal{U}_{\mathrm{hom}}$*, the same equilibrium manifolds* $\mathcal{N}_{\mathrm{car}}, \mathcal{N}_{\mathrm{cur}}, \mathcal{N}_{\mathrm{hom}}$*, the same ambient Hessians* $\mathbb{H}_{\mathrm{car}}, \mathbb{H}_{\mathrm{cur}}, \mathbb{H}_{\mathrm{hom}}$*, the same ambient kernel projectors* $\widetilde{\Pi}_{\mathrm{mic,car}}, \widetilde{\Pi}_{\mathrm{mic,cur}}, \widetilde{\Pi}_{\mathrm{mic,hom}}$*, the same ambient positive gaps* $\rho_{\mathrm{eq,car}}, \rho_{\mathrm{eq,cur}}, \rho_{\mathrm{eq,hom}}$*, the same primitive response operators* $\mathcal{R}_{\mathrm{sel,car}}, \mathcal{R}_{\mathrm{sel,cur}}, \mathcal{R}_{\mathrm{sel,hom}}$*, the same reduced Hessians* $\mathcal{H}_{\mathrm{car}}, \mathcal{H}_{\mathrm{cur}}, \mathcal{H}_{\mathrm{hom}}$*, the same microphysical kernel projectors* $\Pi_{\mathrm{mic,car}}, \Pi_{\mathrm{mic,cur}}, \Pi_{\mathrm{mic,hom}}$*, and the same microphysical coercive gaps* $\delta_{\mathrm{mic,car}}, \delta_{\mathrm{mic,cur}}, \delta_{\mathrm{mic,hom}}$*.*

4. *Consequently the full recorded functional* $\mathscr{F}_{\mathrm{sel}}$ *in* (18) *is preserved without change.*

In particular, the previous charge-generator / continuity-Hessian deeper-origin note is no longer primitive at current scope; it is itself the induced image of the deeper reconfiguration-group / transport-law / equilibrium-reduction substrate layer.

Proof. Items (1), (2), and (3) are the direct contents of Propositions 1, 2, and 3. The functional (18) depends only on the symmetry actions and their Haar projectors, the quadratic compatibility penalties determined by the linearized balance layer, the primitive response operators, and the fixed visible $\mathscr{F}_{\mathrm{proto}}$ slot. Since each of those ingredients is recovered exactly, the functional itself is recovered exactly. □

Theorem 2 (Same exact-preservation and downstream package)**.** *Because the same functional* $\mathscr{F}_{\mathrm{sel}}$ *is recovered, the same downstream package recorded in the current charge-generator / continuity-Hessian deeper-origin note is reproduced without change:*

1. *the same Hessian-layer functional* $\mathscr{F}_{\mathrm{hess}}$*;*

2. *the same microphysical exact-preservation functional* $\mathscr{F}_{\mathrm{micro}}$*;*

3. *the same exact-preservation normal form* $\mathscr{F}_{\mathrm{deep}}$*;*

4. *the same carrier-origin functional* $\mathscr{F}_{\mathrm{car}}[\mathfrak{S}, \mathfrak{R}, \mathfrak{F}]$*;*

5. *the same primitive reservoir* $\mathscr{F}_{\mathrm{prim}}[K, J, F]$*;*

6. *the same microscopic-equilibrium completion reservoir* $\mathscr{F}_{\mathrm{mic}}[H, Y]$*;*

7. *the same zero-free-mode verdict* $\Pi_{\mathrm{free}}\widehat{\Sigma}^{\mathrm{cmp}} = 0$*;*

8. *the same substrate solve* $\widehat{\mathcal{L}}(\widehat{\Sigma}^{\mathrm{cmp}}) = -\widehat{\mathcal{V}}^{\mathrm{res}}$*;*

9. *the same observer solve* $\mathcal{L}_{g^{\mathrm{br}}}(\Sigma^{\mathrm{cmp}}) = -\mathcal{V}^{\mathrm{res}}_{\mu\nu}$*;*

10. *the same* Σ^{cmp} *solve, the same one operative metric, and the same recorded Phase D / E and Phase F gains.*

Proof. Theorem 1 recovers exactly the same functional $\mathscr{F}_{\mathrm{sel}}$ as the one already treated in the current charge-generator / continuity-Hessian deeper-origin note. That note already proves that the stationary unsuppressed reduction of $\mathscr{F}_{\mathrm{sel}}$ gives the same $\mathscr{F}_{\mathrm{hess}}$, the same $\mathscr{F}_{\mathrm{micro}}$, and hence the same downstream chain (19). Since no visible slot or later elimination is changed here, every downstream object listed above is reproduced verbatim. □

Corollary 1 (Recorded gain package is preserved)**.** *Because the same downstream package is recovered, the recorded Phase D / E infrared-spectrum and universal-coupling verdicts, the recorded Phase F controlled GR limit, and the recorded observer-equation removal of the former genuine quartic residual sector remain preserved without modification.*

Proof. This is the direct content of Theorem 2. □

7. Claim-Boundary Verdict

Theorem 3 (Derivation / balance / equilibrium origin verdict)**.** *At the disciplined scope of the present note, the exact positive result is the following.*

1. *The exact-preserving derivation algebras* $\mathfrak{d}_{\mathrm{car}}, \mathfrak{d}_{\mathrm{cur}}, \mathfrak{d}_{\mathrm{hom}}$ *are no longer primitive at current scope.*

2. *They arise as the visible tangent quotients of finite-dimensional exact-preserving reconfiguration groups acting unitarily on the sharper substrate sectors.*

3. *The primitive nonlinear density maps* $\widehat{\mathcal{Q}}_{\mathrm{car}}, \widehat{\mathcal{Q}}_{\mathrm{cur}}, \widehat{\mathcal{Q}}_{\mathrm{hom}}$ *and primitive balance defects* $\widehat{\mathfrak{c}}_{\mathrm{car}}, \widehat{\mathfrak{c}}_{\mathrm{cur}}, \widehat{\mathfrak{c}}_{\mathrm{hom}}$ *are no longer primitive either.*

4. *They arise as the pullbacks of raw substrate transport laws along exact-preserving visible embeddings, and their linearizations again yield the recorded density readouts* $\mathcal{Q}_{\mathrm{car}}, \mathcal{Q}_{\mathrm{cur}}, \mathcal{Q}_{\mathrm{hom}}$*, continuity operators* $\partial_{\mathrm{car}}, \partial_{\mathrm{cur}}, \partial_{\mathrm{hom}}$*, and compatibility-current maps* $\mathfrak{c}_{\mathrm{car}}, \mathfrak{c}_{\mathrm{cur}}, \mathfrak{c}_{\mathrm{hom}}$*.*

5. *The primitive ambient equilibrium energies* $\mathcal{U}_{\mathrm{car}}, \mathcal{U}_{\mathrm{cur}}, \mathcal{U}_{\mathrm{hom}}$ *and equilibrium manifolds* $\mathcal{N}_{\mathrm{car}}, \mathcal{N}_{\mathrm{cur}}, \mathcal{N}_{\mathrm{hom}}$ *are no longer primitive either.*

6. *They arise as reduced pullbacks of deeper exact-preserving equilibrium energies and equilibrium manifolds, so the ambient Hessians are again their second variations, the ambient kernel projectors are again the tangent equilibrium projectors, and the ambient positive gaps are again forced by strict transverse coercivity.*

7. *Therefore the full recorded derivation-law / balance-law / equilibrium-manifold package of the current charge-generator / continuity-Hessian deeper-origin note is recovered without change.*

8. *Consequently the same charge-generator / continuity-Hessian layer, the same* $\mathscr{F}_{\mathrm{sel}}$*, the same exact-preservation normal form, the same carrier-origin functional, the same primitive reservoir, the same microscopic-equilibrium reservoir, the same* $\Pi_{\mathrm{free}} H = 0$ *outcome, the same* Σ^{cmp} *solve, the same substrate and observer solves, the same one operative metric, and the same recorded Phase D / E and Phase F gains remain unchanged.*

9. *Stronger promotion language is still not justified here, because the present note explains only why the previous derivation-law / balance-law / equilibrium-manifold layer arises from a deeper reconfiguration-group / transport-law / equilibrium-reduction substrate layer; it does not yet derive why that new deeper layer is itself unique, final, or inevitable beyond current scope.*

10. *No broader packaging consequence or default side-line continuation follows from the mathematical content of this note alone. The remaining honest burden is one layer deeper again on this same exact-preservation line, or else another comparably concrete observer-equation gain on the same one-metric route.*

Proof. Items (1) and (2) are Proposition 1. Items (3) and (4) are Proposition 2. Items (5) and (6) are Proposition 3. Items (7) and (8) are Theorems 1 and 2 together with Corollary 1. Items (9) and (10) are the present claim-boundary discipline stated in the abstract and introduction. □

Corollary 2 (What remains next)**.** *The primary active burden moves one layer deeper again. It is no longer to justify why the exact-preserving derivation algebras, primitive nonlinear density/balance laws, and primitive equilibrium energies/manifolds of the previous note are admissible at current scope; that step is now recorded. The next honest burden is either to derive or justify why the reconfiguration groups, raw substrate transport layer, and raw equilibrium-reduction geometry introduced here are themselves forced by yet more primitive substrate structure, or else to produce another comparably concrete observer-equation gain on the same one-metric line. No stronger promotion language, no packaging pass, and no default move onto the anchored positive-pair side line are justified unless they directly discharge one of those burdens.*

8. Conclusion

The positive result of this note is that the exact-preserving derivation-law / balance-law / equilibrium-manifold layer is no longer left primitive at present scope. It is recovered from a still deeper reconfiguration-group / transport-law / equilibrium-reduction substrate layer.

At that deeper level, the recorded derivation algebras

$$\mathfrak{d}_{\mathrm{car}}, \qquad \mathfrak{d}_{\mathrm{cur}}, \qquad \mathfrak{d}_{\mathrm{hom}}$$

are the visible tangent quotients of exact-preserving reconfiguration groups acting unitarily on the sharper substrate sectors

$$\mathscr{U}_{\mathrm{car}}, \qquad \mathscr{U}_{\mathrm{cur}}, \qquad \mathscr{U}_{\mathrm{hom}}.$$

Their visible quotient charge Lie algebras, induced generator maps, compact symmetry actions, Haar-average symmetry projectors, and Casimir penalties are therefore recovered again exactly as before.

The primitive density and balance layer is no longer primitive either. The recorded nonlinear density maps

$$\widehat{\mathcal{Q}}_{\mathrm{car}}, \qquad \widehat{\mathcal{Q}}_{\mathrm{cur}}, \qquad \widehat{\mathcal{Q}}_{\mathrm{hom}}$$

and balance defects

$$\widehat{\mathfrak{c}}_{\mathrm{car}}, \qquad \widehat{\mathfrak{c}}_{\mathrm{cur}}, \qquad \widehat{\mathfrak{c}}_{\mathrm{hom}}$$

are the pullbacks of raw substrate transport laws along exact-preserving visible embeddings. Their linearizations therefore recover exactly the same density readouts

$$\mathcal{Q}_{\mathrm{car}}, \qquad \mathcal{Q}_{\mathrm{cur}}, \qquad \mathcal{Q}_{\mathrm{hom}},$$

the same continuity operators

$$\partial_{\mathrm{car}}, \qquad \partial_{\mathrm{cur}}, \qquad \partial_{\mathrm{hom}},$$

and hence the same compatibility-current maps and compatibility operators.

Likewise the primitive ambient equilibrium energies

$$\mathcal{U}_{\mathrm{car}}, \qquad \mathcal{U}_{\mathrm{cur}}, \qquad \mathcal{U}_{\mathrm{hom}}$$

and equilibrium manifolds

$$\mathcal{N}_{\mathrm{car}}, \qquad \mathcal{N}_{\mathrm{cur}}, \qquad \mathcal{N}_{\mathrm{hom}}$$

are reduced pullbacks of deeper equilibrium geometry. Their second variations are again the recorded ambient Hessians

$$\mathbb{H}_{\mathrm{car}}, \qquad \mathbb{H}_{\mathrm{cur}}, \qquad \mathbb{H}_{\mathrm{hom}},$$

their kernels are again exactly the tangent equilibrium directions, and their strict transverse coercivity again forces the same positive-gap structure. Pulling those ambient Hessians back along the fixed exact-preserving readouts therefore recovers the same primitive response operators, the same reduced Hessians, the same microphysical kernel projectors, and the same coercive positive gaps.

Therefore the full recorded functional $\mathscr{F}_{\mathrm{sel}}$ is preserved verbatim. By the immediately preceding note, that exact same functional again yields the same Hessian-layer functional, the same microphysical exact-preservation functional, the same exact-preservation normal form, the same carrier-origin functional, the same primitive reservoir, the same microscopic-equilibrium reservoir, the same $\Pi_{\mathrm{free}} H = 0$ outcome, the same Σ^{cmp} solve, the same one operative metric, and the same recorded Phase D / E and Phase F gains.

The scope remains honest. This note does not claim a final ultraviolet completion or a completed first-principles GR derivation. Its exact positive result is narrower and precisely the one now needed: the previous derivation-law / balance-law / equilibrium-manifold layer is itself no longer primitive at current scope, but arises from a still deeper reconfiguration-group / transport-law / equilibrium-reduction substrate layer while preserving the same downstream package. The next

honest burden is one layer deeper again: whether that new deeper layer can itself be derived from yet more primitive substrate structure, or else superseded by another equally concrete observer-equation gain on the same one-metric line.